Public Policy 2026-2027

Lecture 7: Market Failures (Continued) & Fairness

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Stockholm School of Economics in Riga

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The Public Goods Typology

Non-excludability:

My having access to the good means you have access to it as well.

Non-rivalry:

My using the good does not reduce your access to the good. (= Zero marginal social costs of consumption, but positive marginal costs of production)

Congestability:

My using (a certain quantity) of the good exhibits negative externalities of consumption.

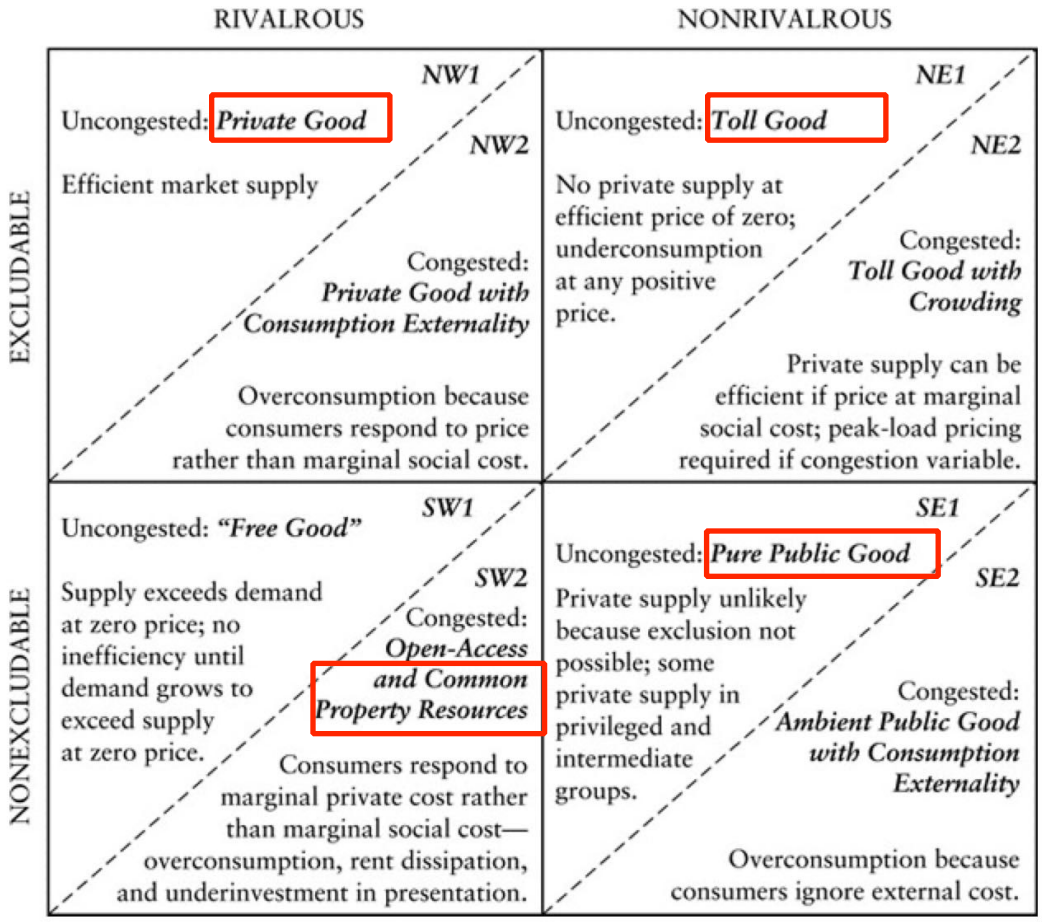
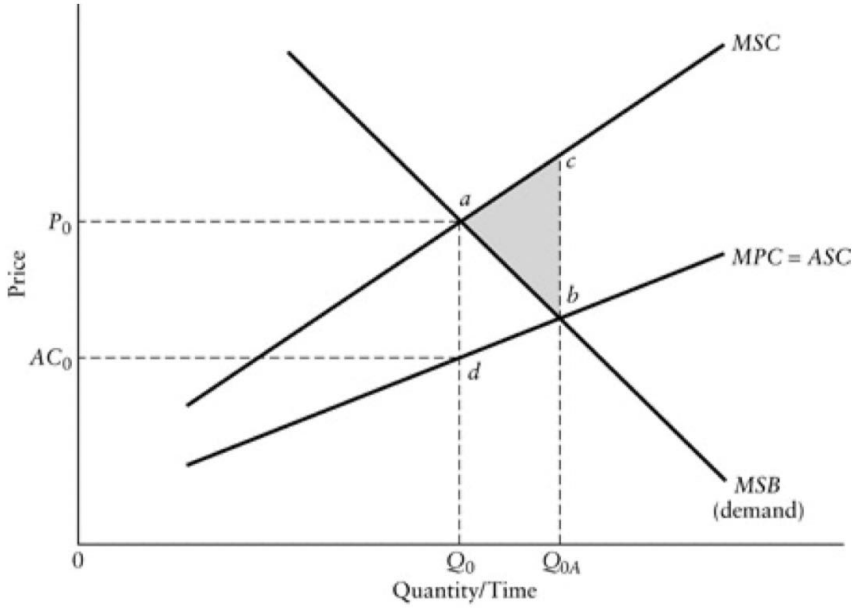


Figure 5.2 A Classification of Goods: Private and Public

Open-Access (or Common Pool) Goods

- Non-excludable ownership, rivalrous in consumption.
- Fish stock, irrigation systems, oil, grassland, internet backbone, etc.
- "Tragedy of the Commons:" Non-excludability leads to Pareto inefficient overconsumption and/or underinvestment — and ultimately — depletion of the good.
- Scarcity rent: If Q is limited at the Pareto-efficient level, who gains, who loses?

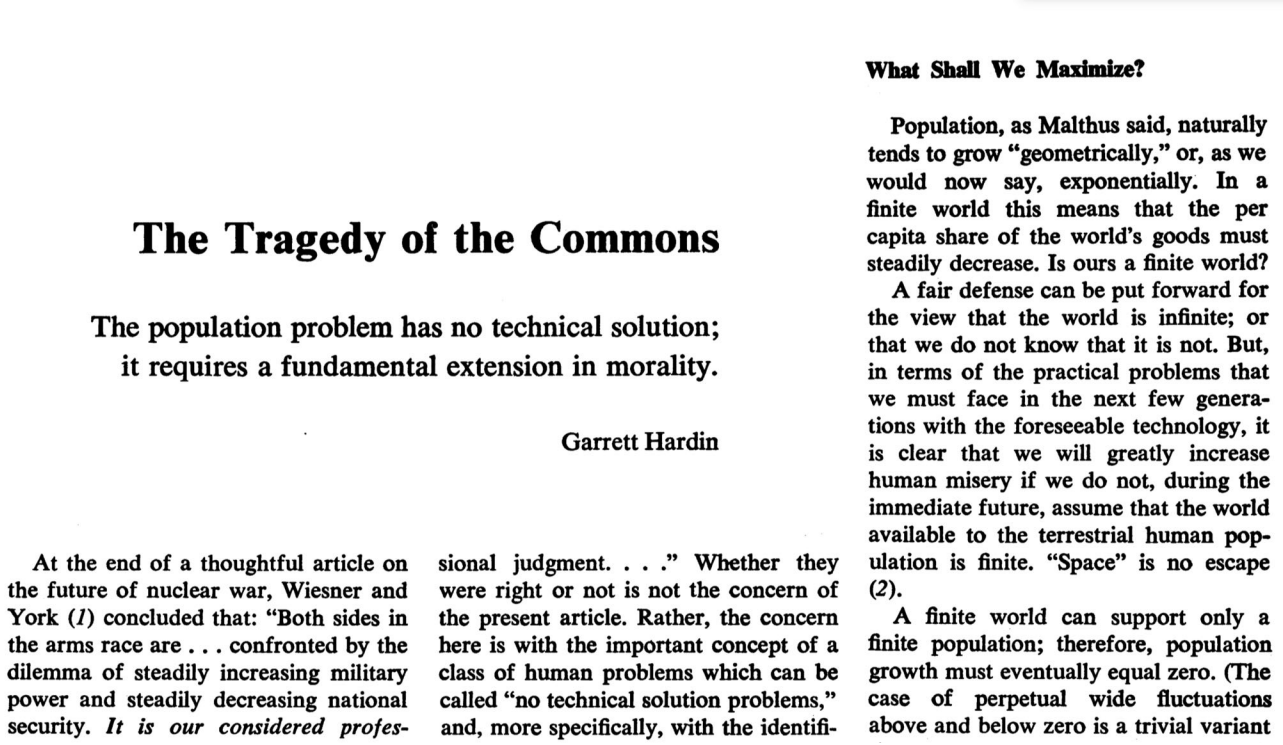


MPC = Marginal private cost
MSC = Marginal social cost
ASC = Average social cost

Socially optimal level of consumption: Q_0 (*MSB* = *MSC*)
Open access resource level of consumption: Q_{0A}
Deadweight loss from overconsumption: *abc*

Overconsumption of Open-Access Resources

The “Tragedy of the Commons”



The Tragedy of the Commons

The population problem has no technical solution; it requires a fundamental extension in morality.

Garrett Hardin

At the end of a thoughtful article on the future of nuclear war, Wiesner and York (*1*) concluded that: “Both sides in the arms race are . . . confronted by the dilemma of steadily increasing military power and steadily decreasing national security. *It is our considered professional judgment. . . .*” Whether they were right or not is not the concern of the present article. Rather, the concern here is with the important concept of a class of human problems which can be called “no technical solution problems,” and, more specifically, with the identifi-

What Shall We Maximize?

Population, as Malthus said, naturally tends to grow “geometrically,” or, as we would now say, exponentially. In a finite world this means that the per capita share of the world’s goods must steadily decrease. Is ours a finite world?

A fair defense can be put forward for the view that the world is infinite; or that we do not know that it is not. But, in terms of the practical problems that we must face in the next few generations with the foreseeable technology, it is clear that we will greatly increase human misery if we do not, during the immediate future, assume that the world available to the terrestrial human population is finite. “Space” is no escape (*2*).

A finite world can support only a finite population; therefore, population growth must eventually equal zero. (The case of perpetual wide fluctuations above and below zero is a trivial variant

	50 Cows	100 Cows
50 Cows	5,5	3,6
100 Cows	6,3	4*,4*

Garrett Hardin (1968) “The Tragedy of the Commons.” *Science* 162(3859), pp. 1243-1248.

Implications of Market Failures for Public Policy

How to Get People to Internalise their Externalities?

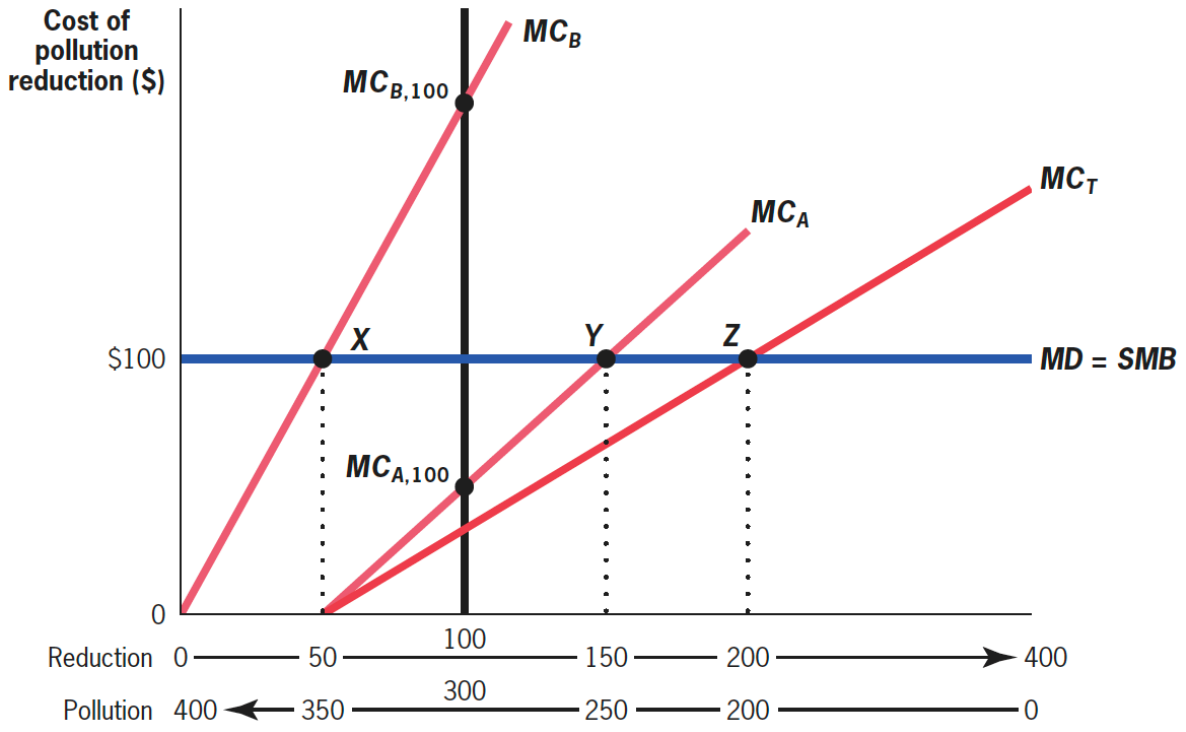
1. The force of persuasion.
2. Pigouvian subsidies and taxes
3. Direct regulation
4. Coasian bargaining
5. Decentralised self-governance

The Force of Persuasion

- Why can’t policymakers simply explain to people that internalising their externalities would make everyone better off?
- Remember the Prisoners’ Dilemma: even if I think that not everybody in my society is a selfish crook, it is still a best response for me to shirk.
- Even if everyone else is chipping in, it is still not in my interest to join…
- Some more intrusive kind of policy action is called for!

Tax vs. Regulatory Solutions to Externalities

- **Price**-based instruments:
  - Punitive (Pigouvian) tax equal to the negative externality (MD).
  - Generally efficient (second-best).
- **Direct regulation** without trade:
  - Government forces firms to produce the socially efficient quantity.
  - Generally not efficient.
- **Direct regulation** with trade ("cap and trade" schemes):
  - Government issues permits that allow a certain amount of pollution and let the firms trade.
  - Efficient, even if MC uncertain!



Second-best policy:
The policy that maximises social welfare, taking into consideration all the various effects of the policy.

A Market in Externalities

The “Coase Theorem”

- Are externalities really outside the market mechanism?
- The “Coase Theorem:”
  - *The Problem of Social Cost* (1960)
  - Part I: The party causing the externality and the party affected by the externality will voluntarily bargain about any issue over which they conflict.
  - Part II: The outcome of the bargaining between the parties will be Pareto-efficient, regardless of the initial distribution of property rights.
- George J. Stigler (NP 1982): “How so fine an economist could make such an obvious mistake.”



Ronald H. Coase 1910-2013

Crop Farmer vs. Cattle Rancher

Coase's original example

		Property rights assignment	
		Farmer has property right	Rancher has property right
Social optimum	Not Ranching (B<C)	Rancher moves	Farmer pays Rancher between B and C to stop or reduce ranching
	Ranching (B>C)	Rancher pays Farmer between C and B to accept damage or move	Farmer moves





Ranchers impose a negative externality on crop farmers. Coasean bargaining predicts that the efficient outcome is the same in each row, whichever column you're in!

Implications of the Coase Theorem

- Coase (1960) makes two crucial assumptions:
  - **Property rights are clearly assigned:** economic agents exactly know who has the right to pollute and who not.
  - **Transaction costs are sufficiently low:** costs of identifying stakeholders, negotiating contracts, monitoring for compliance, sanction mechanisms…
- Conclusion of the practitioner:



- Conclusion of the economist: the market mechanism informs institutional design!

Implications of the Coase Theorem

- Coasean bargaining might be an effective internalization strategy in the presence of small-scale, localized externalities.
- Unsuitable for large-scaled externalities (global warming etc.).
- Coase hope was that governments…
  - …improve the quality of their institutions in view of lowering transaction costs.
  - …create opportunities for the creation of new markets for trading externalities.
  - Better define the role of the government in solving externality problems.

Problems With Coasean Bargaining in the Real World

1. Unruly Passenger 1 (claimed right to recline seat)
2. Unruly Passenger 2 (claimed right to legroom)

Post Nation

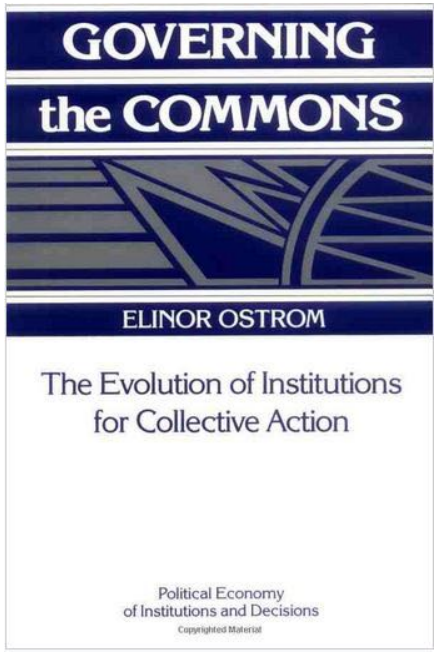
United flight diverted after passengers fight over legroom

By Abby Phillip August 26, 2014



Self-Governance

- The “Tragedy of Commons” is neither universal nor unavoidable!
- Collective Self-Governance can overcome the “Tragedy”
- State ownership and privatization are not the only solutions to the free rider problem
- Hardin’s “Tragedy” happens when
  - resource stakeholders are anonymous
  - get no feedback on individual actions
  - do/can not communicate



Ostrom, Elinor. 1990. *Governing the Commons: The Evolutions of Institutions for Collective Action*. Cambridge UP.



Elinor Ostrom (1933-2012)

Self-Governance: Ostrom's Eight Design Principles

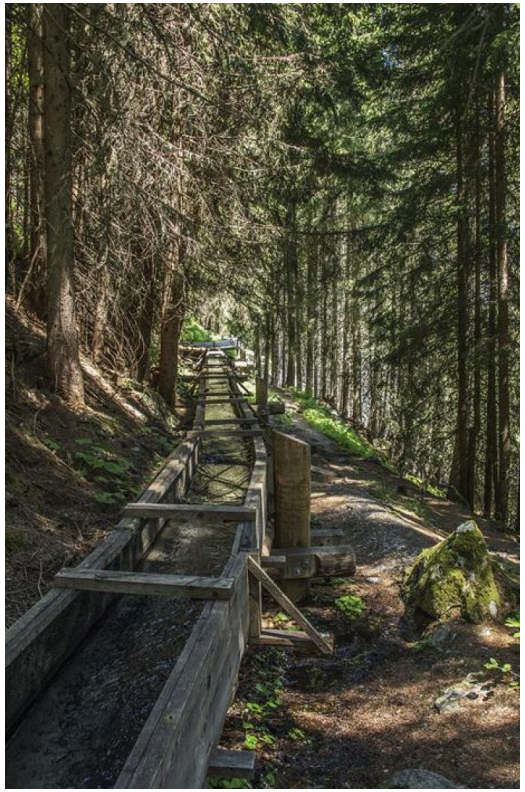
Ostrom's institutional design principles	Description
1. Clear boundaries	Individuals who have the right to use the resource are clearly defined
2. Congruence with local conditions	Rules restricting time, place, technology, & quantity of resource use are adapted to local conditions
3. Collective-choice arrangements	Individuals affected by resource-related rules can participate in modifying them
4. Monitoring	Resource conditions & use are monitored by the users themselves or by people accountable to users
5. Graduated sanctions	Credible threats & enforcement of penalties exist for users who violate resource-related rules
6. Conflict-resolution mechanisms	Users & officials have rapid & low-cost access to mechanisms of resolving conflicts
7. Recognition of right to organization	The rights of users to devise their own institutions are not challenged by government authorities
8. Multi-level governance	Appropriation, provision, monitoring, conflict-resolution are organised in multiple, nested layers

Successful Common Pool Self-Governance over Centuries: Land and Irrigation Regimes





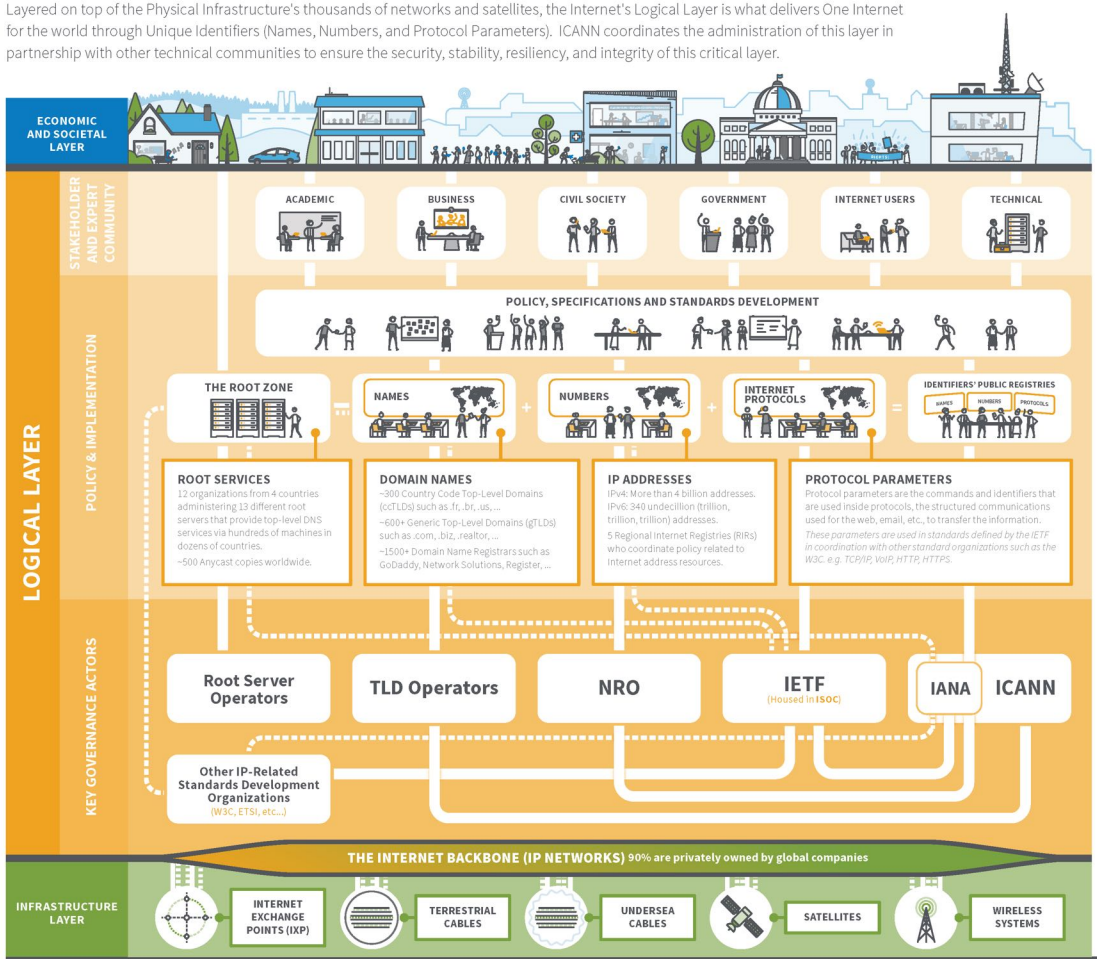




The Global Governance of the Internet

THE LOGICAL LAYER OF DIGITAL GOVERNANCE

Layered on top of the Physical Infrastructure's thousands of networks and satellites, the Internet's Logical Layer is what delivers One Internet for the world through Unique Identifiers (Names, Numbers, and Protocol Parameters). ICANN coordinates the administration of this layer in partnership with other technical communities to ensure the security, stability, resiliency, and integrity of this critical layer.



TECHNICAL OPERATIONS

The technical operating community is made up of multiple independent actors bound by common principles and mutual commitments that ensure the security and stability of the Internet Infrastructure. Each actor's community develops policies and standards in an open, inclusive, and consensus-based approach.

KEY GOVERNANCE ACTORS

ICANN *Internet Corporation for Assigned Names and Numbers*
Helps coordinate the Internet's systems of unique identifiers including domain names and IP addresses, as well as manages the IETF's protocol parameter registries.
www.icann.org

IANA, the Internet Assigned Numbers Authority, is a set of functions housed and operated within ICANN. It acts as the top-level allocator for blocks of IP addresses and AS numbers, proposes creation of and changes to DNS top-level domains, and manages lists of unique identifiers used in Internet protocols.
www.iana.org

IETF *Internet Engineering Task Force*
Develops and promotes a wide range of Internet standards dealing in particular with standards of the Internet protocol suite. Their technical documents influence the way people design, use, and manage the Internet. The IETF operates under the Internet Society (ISOC) with architectural oversight provided by the Internet Architecture Board (IAB).
www.ietf.org

ISO *International Organization for Standardization*
Standardizes, among many other things, the official names and postal codes of countries, dependent territories, special areas of geographic significance.
www.iso.org

NRO *Number Resource Organization*
A coordinating body for the five Regional Internet Registries (RIRs). The RIRs manage the distribution of IP addresses and Autonomous System Numbers in their regions of the world.
www.nro.net
AFRINIC *www.afrinic.net*
APNIC *www.apnic.net*
ARIN *www.arin.net*
LACNIC *www.lacnic.net*
RIPE NCC *www.ripe.net*

TLD Operators *Top Level Domain Operators*
Organizations which have been assigned the management of Top-Level Domains such as: Generic TLDs (.com, .edu, .info, .name etc ...), Country Code TLDs (.fr, .us, .gh, .cn etc...) and non-ASCII alphabet TLDs (in language such as Chinese, Korean, Arabic, Russian, French etc...) —among others.

Root Server Operators
12 independent organisations operate the 13 authoritative name servers (A through M) that serve the Domain Name System (DNS) root zone. The name servers are a network of hundreds of physical servers located in many countries around the world.
www.root-servers.org

W3C
The World Wide Web Consortium (W3C) is an international community where Member organizations, a full-time staff, and the public work together to develop Web standards. W3C's mission is to lead the Web to its full potential.
www.w3.org

STAKEHOLDER AND EXPERT COMMUNITY

Academic
- Institutions of higher learning
- Academic thought leaders
- Professors & students

Business
- Private-sector companies from across industries
- Industry and trade associations

Civil Society
- International organizations
- Non-governmental organizations
- Non-profit organizations
- Think Tanks

Government
- National governments
- Distinct economies recognized in international fora
- Multinational governmental and treaty organizations
- Intergovernmental organizations
- Public authorities (with a direct interest in global Internet Governance)

Internet Users
- Private citizens interested in regional or global Internet Governance

Technical
- Internet engineers
- Computer engineers
- Software developers
- Network operators

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Trading Off Efficiency and Distributional Concerns (“Fairness”)

Individual Inequality Aversion

Individual preferences for inequality

- Disadvantageous inequality
- Advantageous inequality

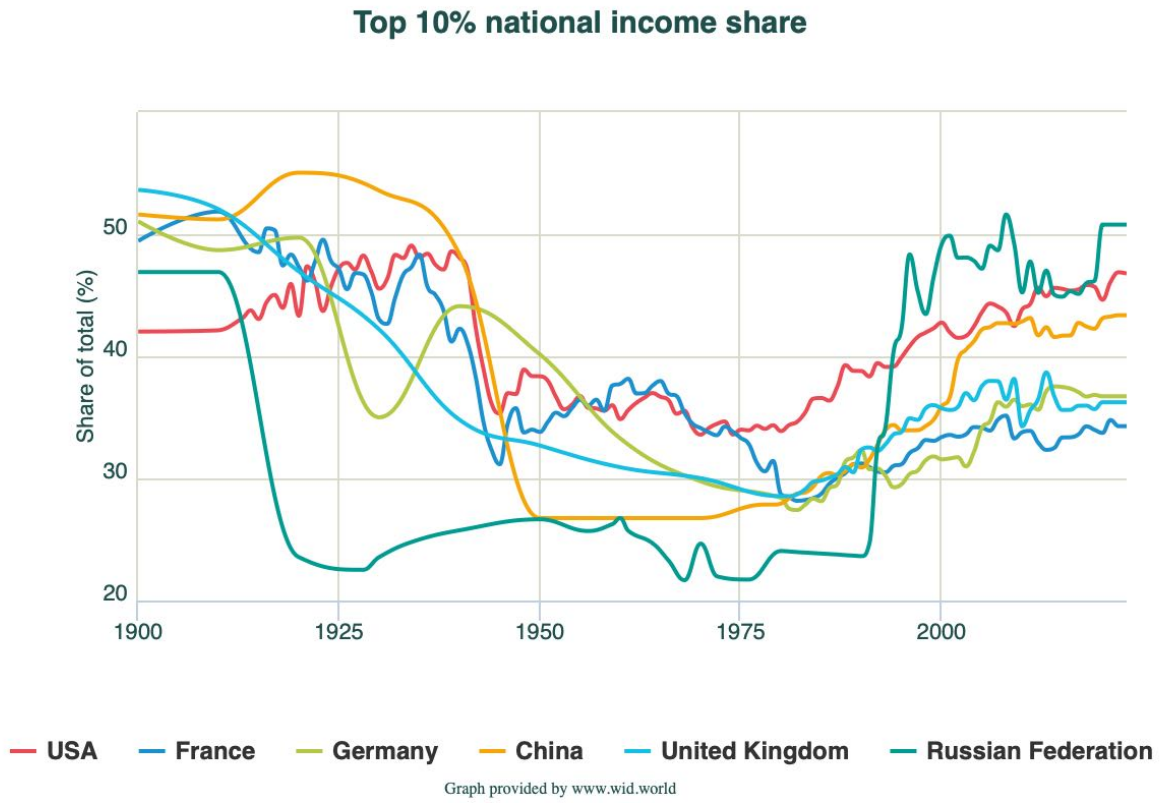
$$U_i(\Pi_i, \Pi_j) = \Pi_i - \alpha \max(\Pi_j - \Pi_i, 0) - \delta \max(\Pi_i - \Pi_j, 0)$$

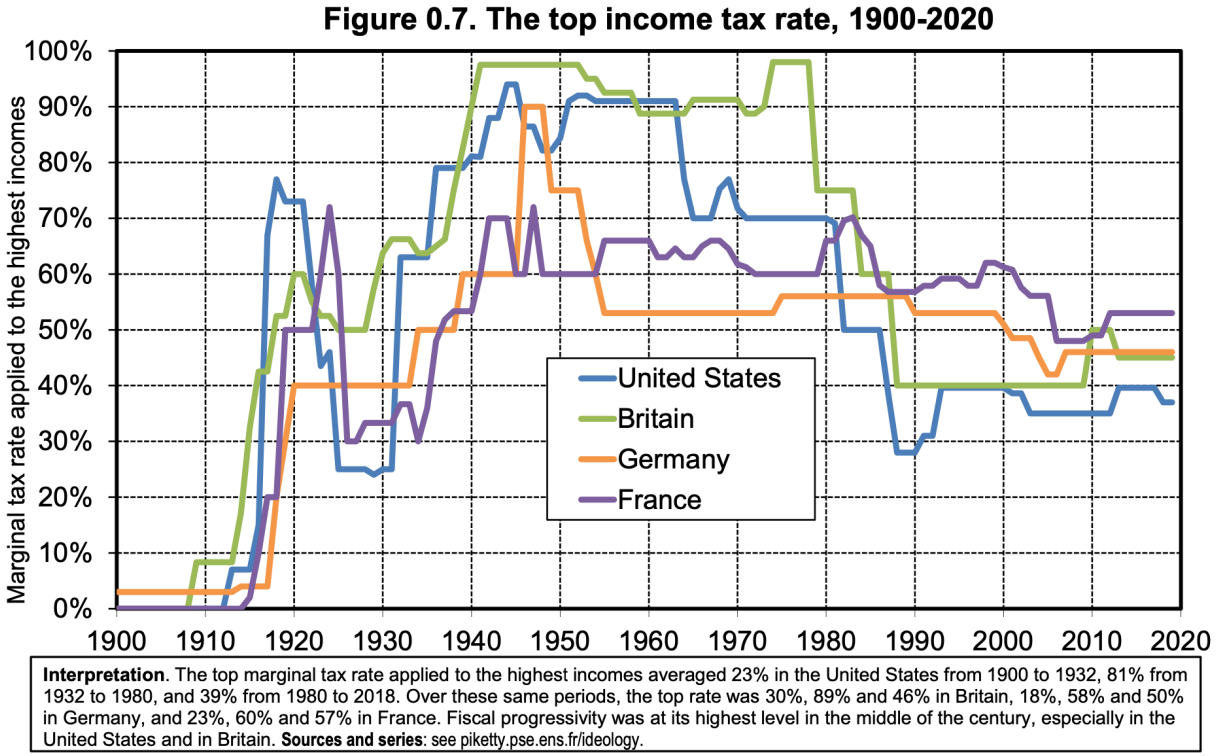
$$\alpha \in [0, 1]$$

$$\delta < \alpha$$

Dividing the Income Pie, 1900-2020s

Social preferences for inequality

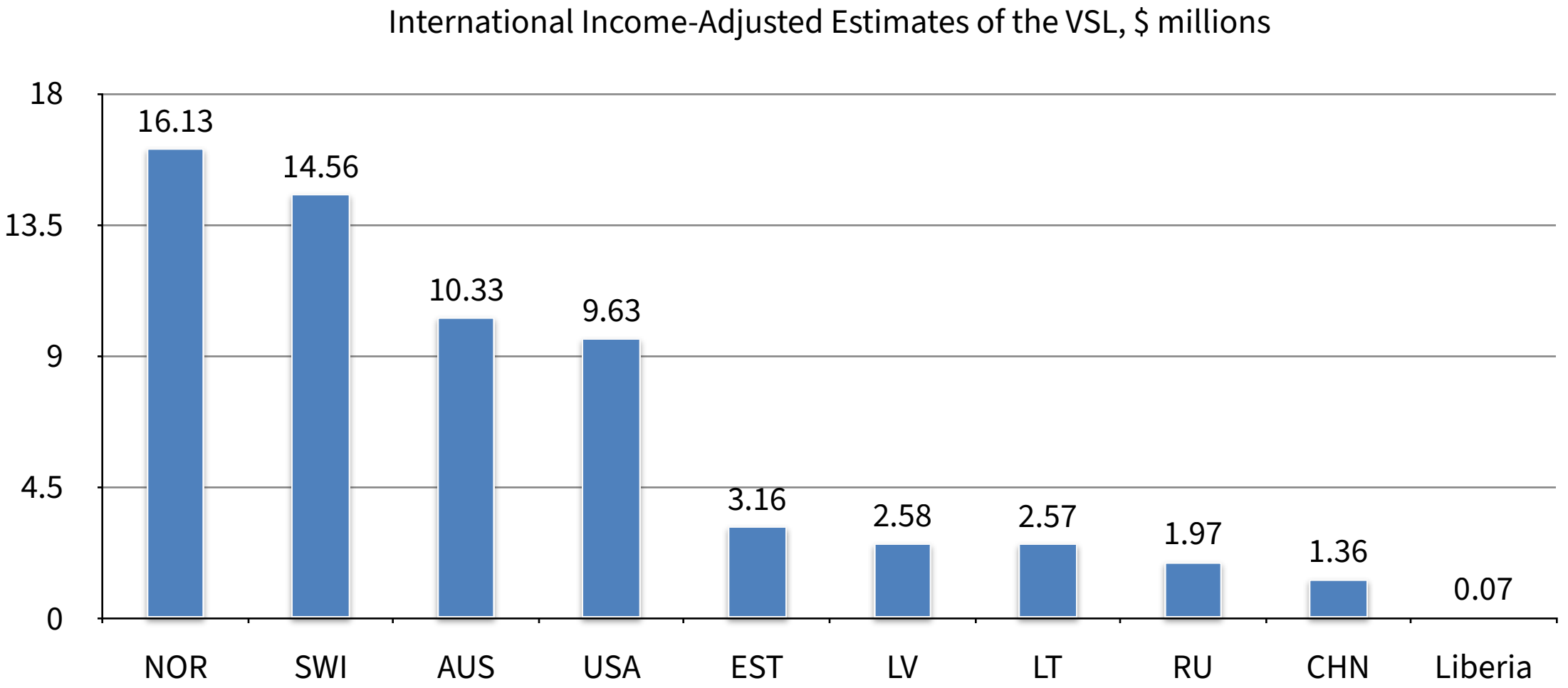




Data from the World Inequality Database Project

The Value of a Statistical Life (VSL)

Social preferences for inequality



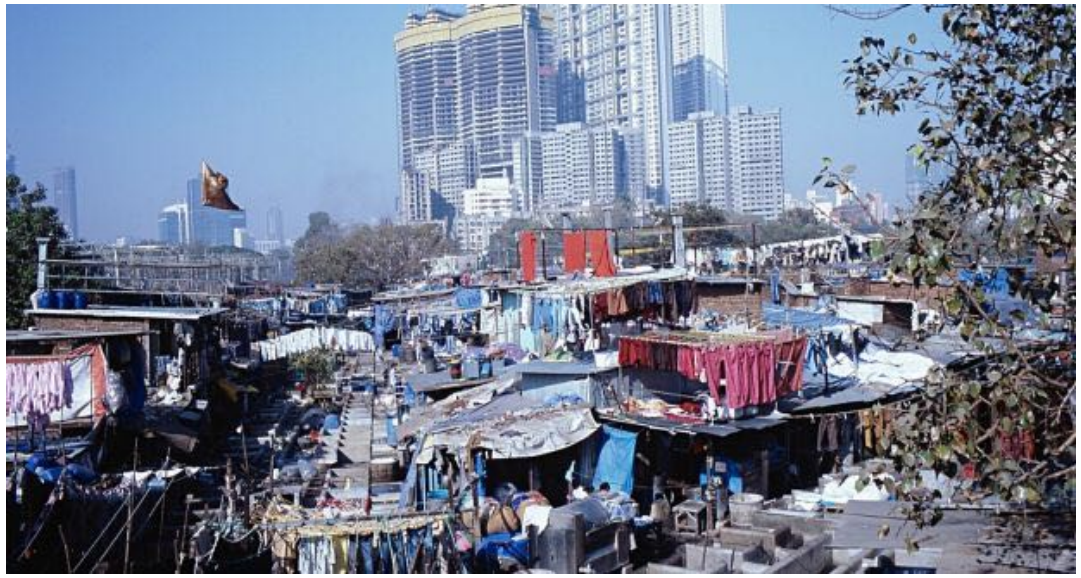
Kip Viscusi & Clayton J. Masterman. 2017. "Income Elasticities and Global Values of a Statistical Life." *Journal of Benefit-Cost Analysis* 8(2). 226-250

Social Inequality Aversion

Social preferences for inequality

"An economy can be optimal in this sense even when some people are rolling in luxury and others are near starvation as long as the starvers cannot be made better off without cutting into the pleasures of the rich... In short, a society or an economy can be Pareto-optimal and still be perfectly disgusting."

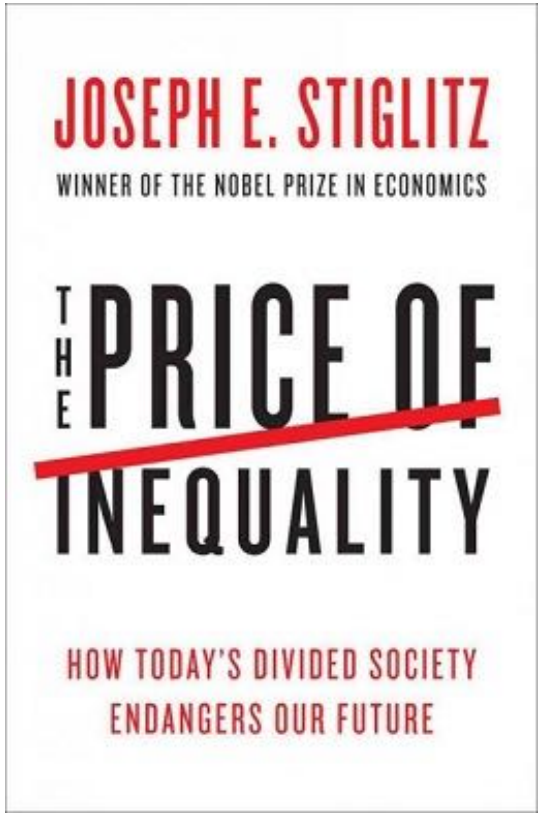
—Amartya Sen. 1970. *Collective Choice and Social Welfare*, p.22



Dhobi Ghat in Mumbai, India. © Marcos Freire/EyeEm/Getty Images

What Reasons Does Society Have to Care About Fairness?

- On some moral accounts, undeserved inequalities in welfare are unjust per se.
  - e.g. luck egalitarianism.
  - Main difficulty: is is always appropriate / feasible for the state to distinguish deserved from undeserved disadvantages?
- On other moral accounts, inequality has unacceptable consequences!
  - e.g. rule utilitarianism (contrast with Bentham's act utilitarianism).
  - Destruction of social fabric/ cooperation/ stability.
  - Distorts allocative efficiency of markets (see e.g. the work by Joe Stiglitz)
  - Deteriorates self-respect.



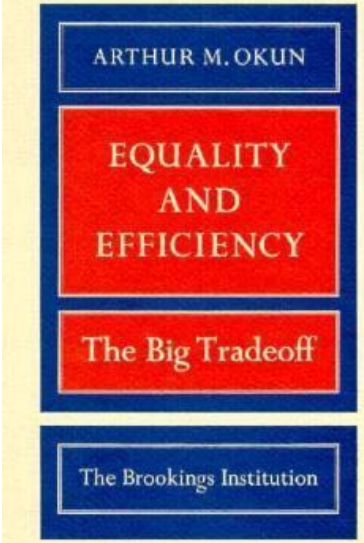
2012

Arthur Okun's "Leaky Bucket"

At what rate should society trade off equality against efficiency?

- Imagine the government taking money from wealthy citizens through taxes and transferring it to poor citizens
- Inequality-averse people have a preference for more income redistribution. The trouble: transferring €1 from a rich person to a poor person resembles a "leaky bucket":
  - Administrative costs of the transfer instrument.
  - DWL from taxation.
  - Moral hazard: DWL from reduced labor supply.
- At what level of leakage are we no longer willing to undertake this transfer?





1975

Arthur Okun's "Leaky Bucket"

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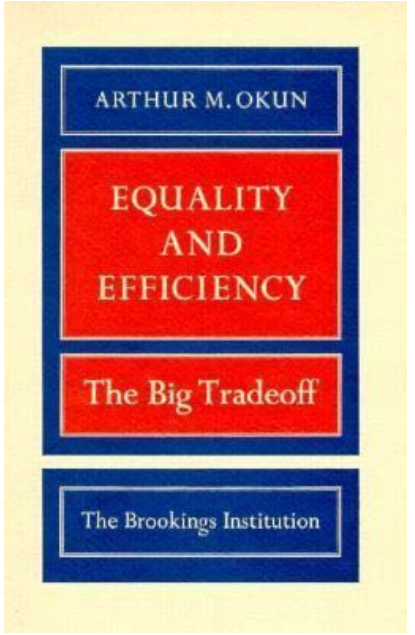
Assuming an isoelastic social valuation of money:

$$(1-L)\,y_P^{-\eta} = y_R^{-\eta}$$

$$L^* = 1 - \left(\frac{y_P}{y_R}\right)^{\eta}$$

y_i = income

η = inequality aversion parameter (= the elasticity of the social marginal utility of income)



1975

Society's Inequality Aversion Parameter

Elasticity of the social marginal utility of income

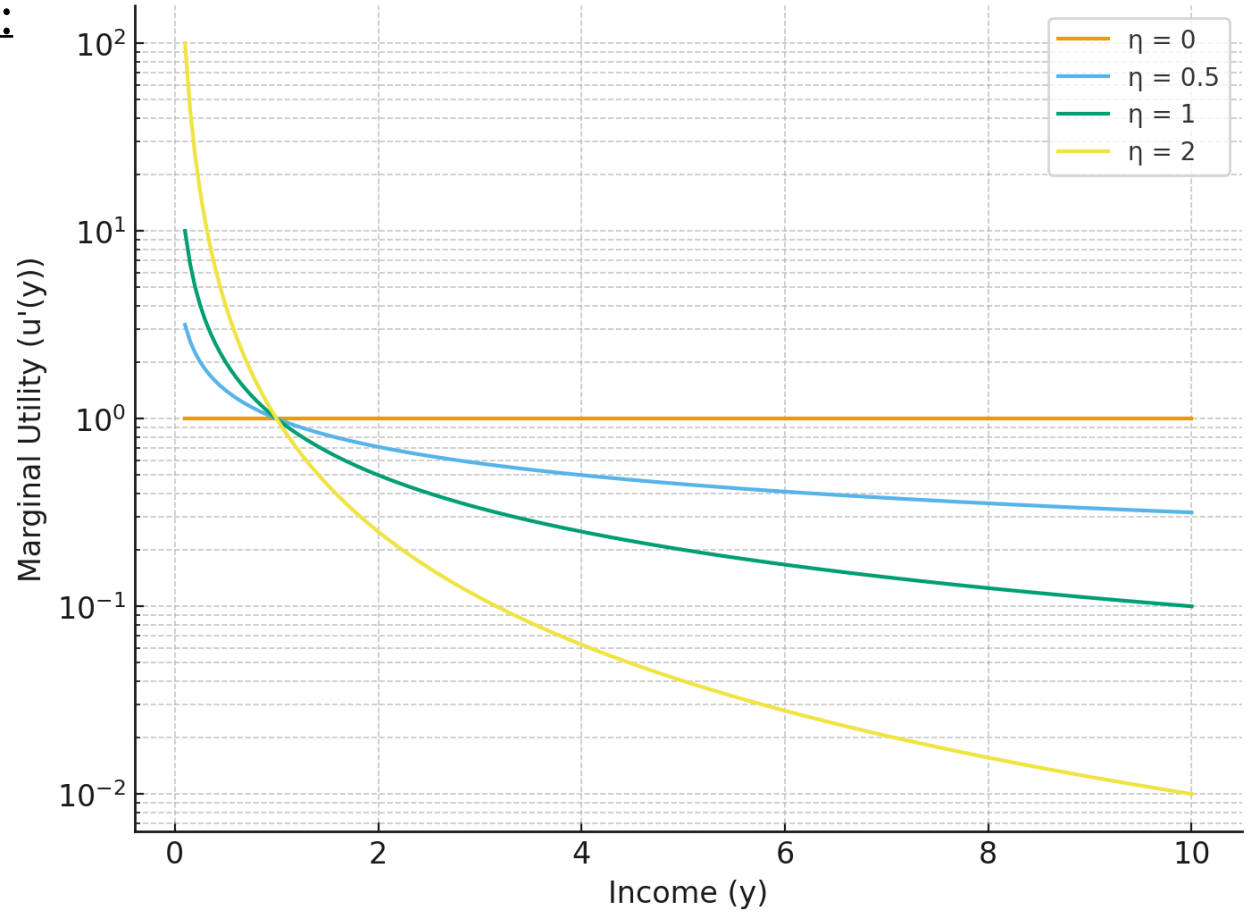
Isoelastic social welfare functions (generic form):

$$W = \sum_i \frac{y_i^{1-\eta}}{1-\eta}$$

for $\eta \neq 1$, and

$$W = \sum_i \ln(y_i)$$

for $\eta = 1$.

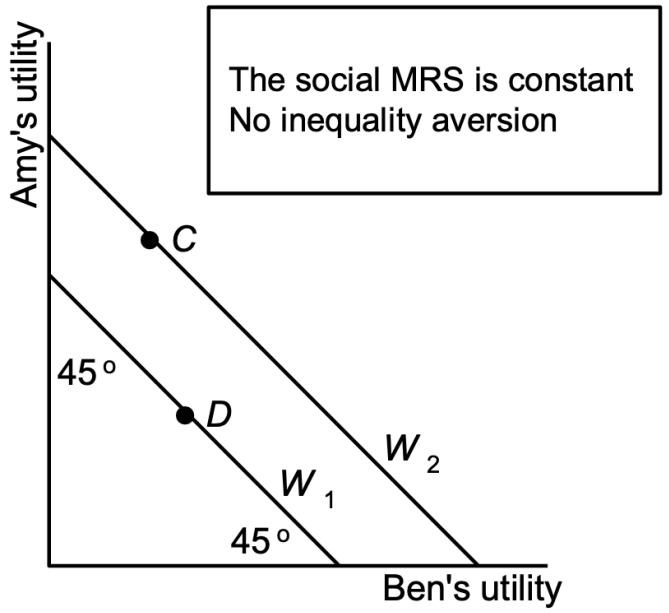


The higher the value of η, the faster the rate at which marginal social utility — **the marginal social value of an extra euro** — falls!

Three Types of Social Indifference Curves

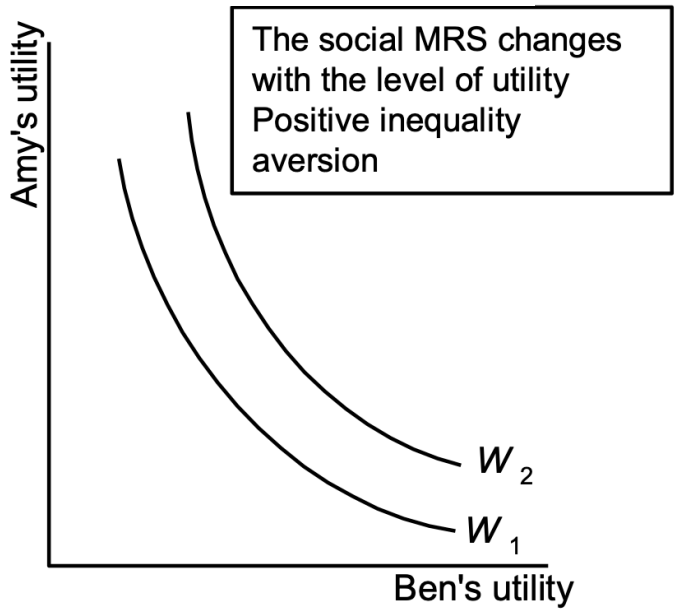
Social utility trade-offs: Bentham, Atksinson/ prioritarian, and Rawls

$$W = \sum_i y_i$$



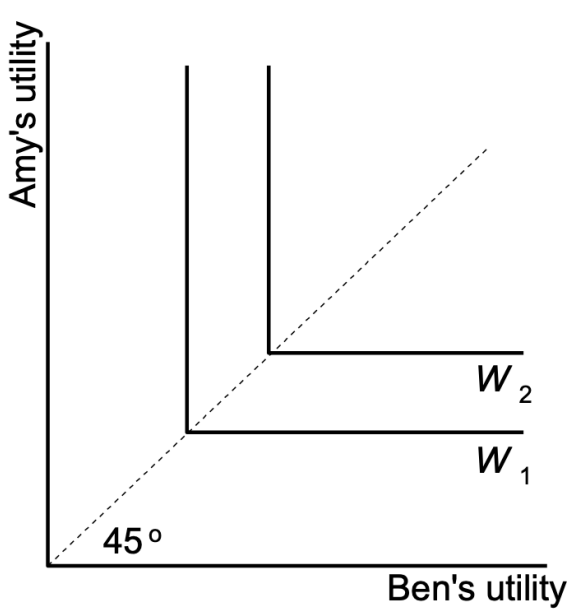
(a) Additive utilitarian SWF

$$W = \sum_i \frac{y_i^{1-\eta}}{1-\eta}$$



(b) Weighted additive utilities SWF

$$W = \min_i y_i$$



(c) Rawlsian SWF

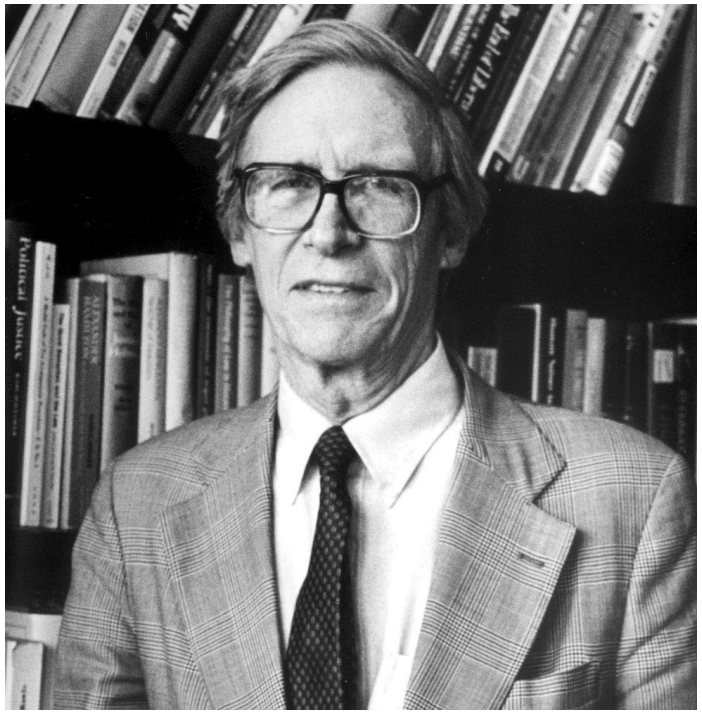
η = 0: complete inequality neutrality

η > 0: some inequality aversion

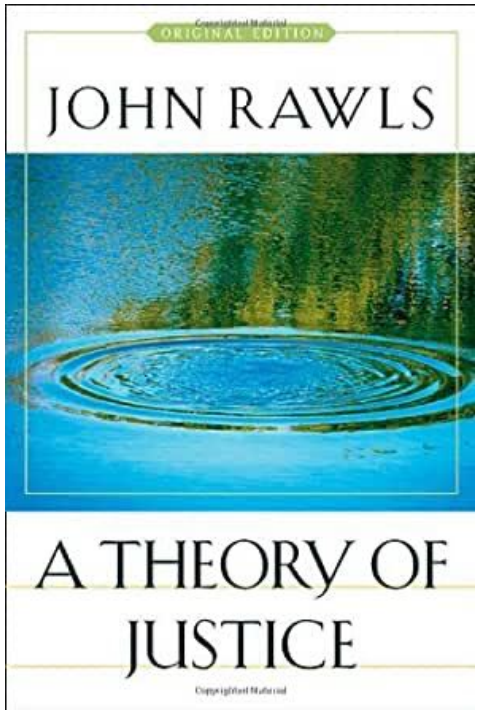
η = ∞: total inequality aversion

John Rawls's *Theory of Justice* (1971)

Extreme inequality aversion



John Rawls (1921-2002)



(1971, rev. 1999)

- ToJ = the seminal statement of post-war liberal egalitarianism.
- ~2.500 books & articles written on ToJ 10y after its publication.
- Lays out an account of a just society with a strong aversion against unequal distributions of primary resources.
- One of the most influential philosophical justifications of redistributive policies to date.

Rawls' Two Principles of Justice

What is a just society? How much economic inequality is tolerable?

1. Each person is to have an equal right to the most extensive total system of **equal basic liberties** compatible with a similar system of liberty for all.
2. Social and economic inequalities are to be arranged so that they are both:
   2.1. attached to offices and positions open to all under conditions of fair **equality of opportunity**, and
   2.2. to the greatest benefit of the **least advantaged**.

The "Difference Principle"

What Society Would You Choose to Be Reborn In?

Rawls' bet: under the veil, rational parties would choose the basic structure (think: constitution) in which your hypothetical worst case is better than your hypothetical worst case in any other alternative economic structure.

- **Maximin**: "maximise the minimal (worst possible) outcome"
- Why not choose average utility?

Society	Red	Blue	Green	Average Income
A	100	100	100	100
B	150	100	50	100
C	600	600	99	433
D	1096	102	101	433

Policy Implications of Inequality Aversion

Four Welfare State Regimes

Welfare regime	Ideological goal	Main provider of welfare	Workforce emancipation from the labour market	Degree of stratification	Allocation of social rights	Involvement of the state in providing welfare	Historical political coalitions	Archetypical examples
Liberal	To promote individualism and **traditional work ethic** through the labour market.	The labour market (individual)	**Lowest** – welfare benefits are directly tied to participation in the labour market.	**Moderate**, with class dualism between the poor state welfare recipients, and the majority who secure insurance via the private sector.	Market differentiated – based on individual needs and means-tests.	**Residual**, providing only a "safety net" at the margin.	Right-wing dominant	United States, Canada, Australia.
Corporatist	To promote traditional values such as **familyhood**, preserve class differentials and legitimize the central authority.	Kinship (family)	**Low** – welfare benefits are tied to employment, although to a lesser extent than in the liberal model.	**Highest**, due to prioritizing the preservation of class politics.	Based on class strata and tied to contributions.	**Central** – private welfare providers are crowded out.	Conservative, traditional, Catholic influence	Austria, France, Germany, Italy.
Social-democratic	To promote **equality** of rights and benefits enjoyed by all members of society.	State	**High**, due to the presence of labour unions and welfare benefits being largely independent of employment, but regime still reliant on production through passive labour market policies.	**Lowest**, with the focus being equality of rights, benefits and status.	Universal, egalitarian access to rights.	**Main provider**	Left-wing dominant	Sweden, Norway, Denmark.

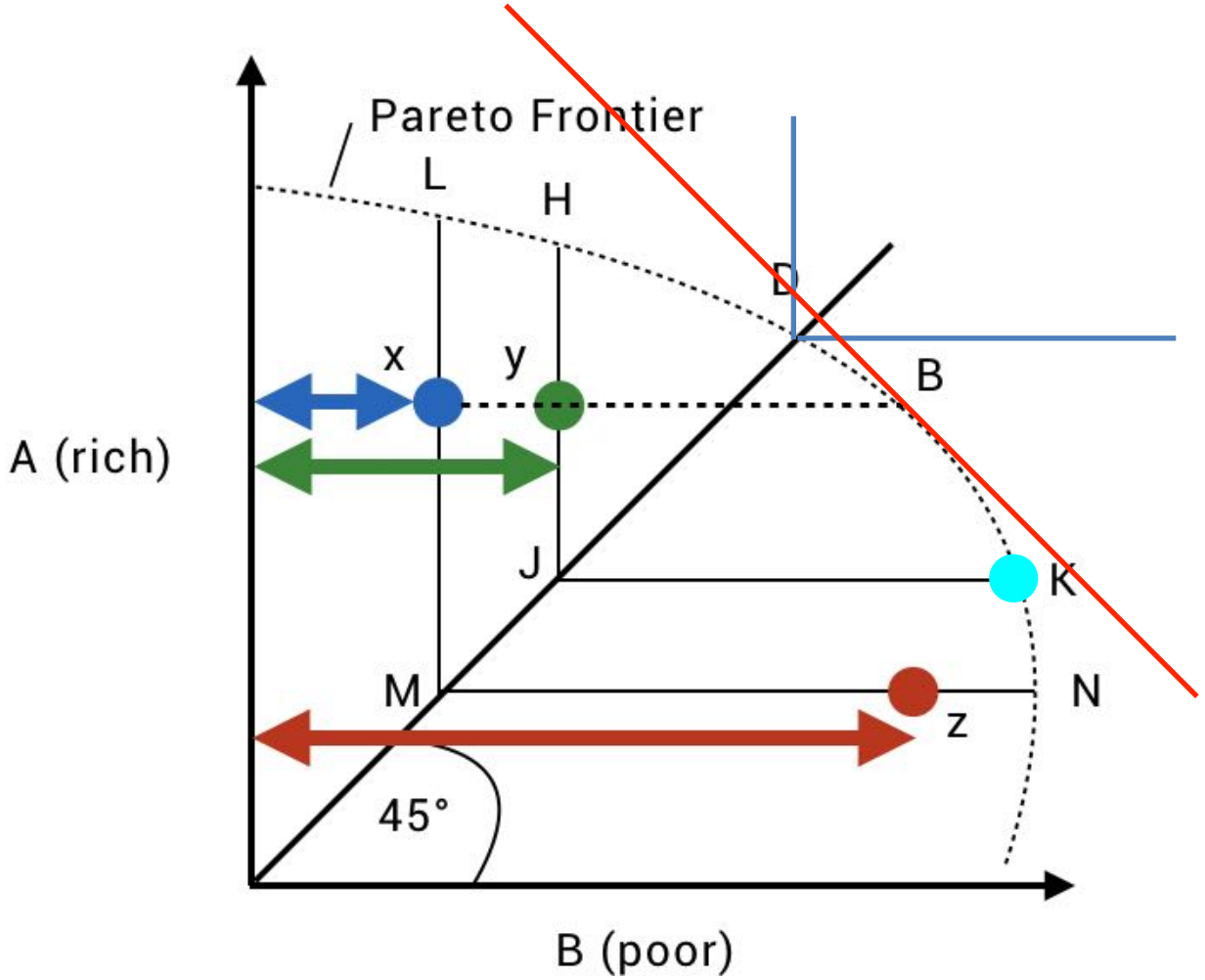
Adjusted from Lukas Latauskas & Eriks Sniedze (2020). *Typology of the Latvian Welfare State*. SSE Riga BSc Thesis.

Bentham, Pareto, and Rawls Compared

x = starting point

What to say about changes from:

- x to H or D?
- x to y?
- x to J?
- x to K?
- x to z?



Jointly Increasing Efficiency and Fairness?

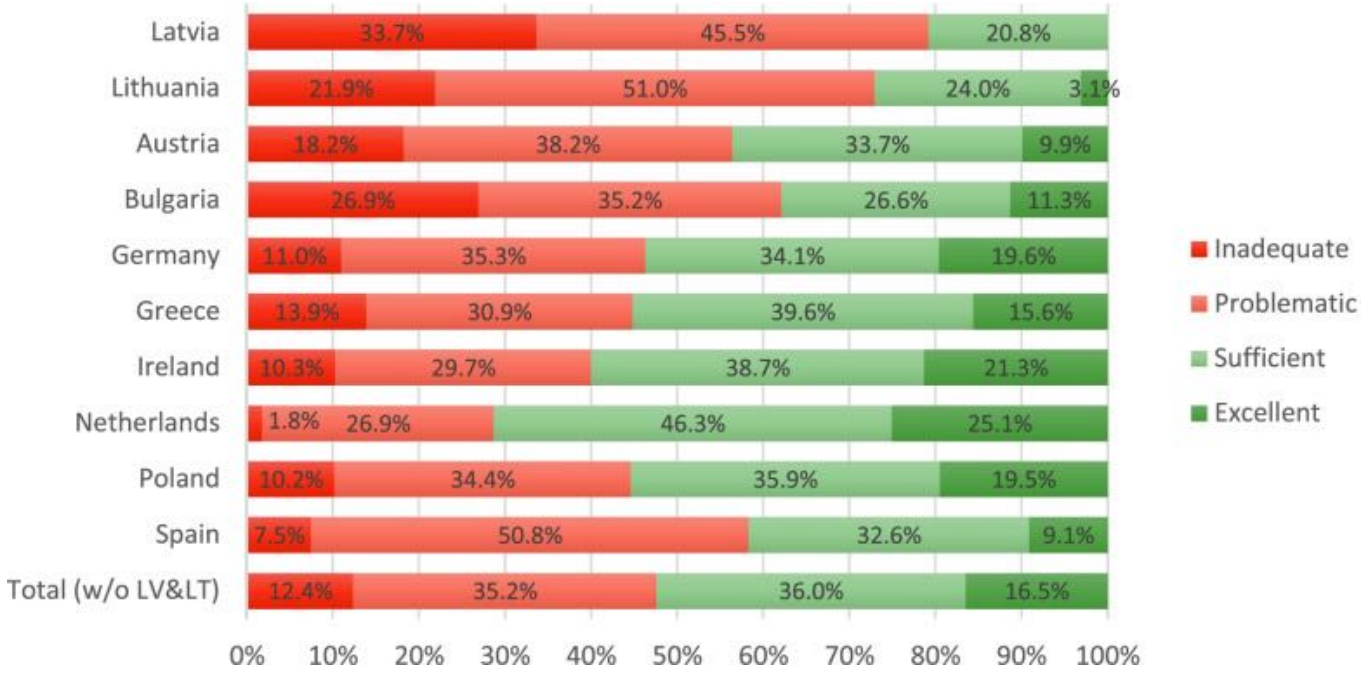
Consider the efficiency and fairness impact of the following policies:

- Increasing the effectiveness of antitrust laws.
- Tightening regulations on management compensation in the financial sector.
- Restricting (or increasing the transparency of) lobbying activities within legislatures.
- Reforming tax policies that tax capital gains at a low rate.
- Improving educational performance in public schools.
- Improving health literacy.

Example: Health Literacy

"Cognitive and social skills which determine the motivation and ability of individuals to gain access to, understand and use information in ways which promote and maintain good health."

— World Health Organization



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Example: Health Literacy

Relationship with health costs and health equity

Efficiency cost of low health literacy:

- Low health literacy is empirically associated with:
  - more visits to the general practitioner.
  - increased rates of hospitalisations (in-patient care).
  - longer stays in hospitals.
  - higher rates of emergency service use.
  - under-utilisation of preventive healthcare services.

Equity cost of low health literacy:

- Health equity: “the attainment of the highest level of health for all people” (WHO).
- Provisional evidence (from the US) showing that health literacy mediates socio-economic determinants of health, particularly education, age, income, and race/ethnicity.
- Hypothesis: health literacy critically influences how social disadvantages translate into disparate health outcomes.

Thank You